

Task 1:

Home Wi-Fi Setup Network Devices Comparison Table

Objective: Compare core network devices (Switch, Router, Hub, Firewall, and Wireless Access Point) used in home and small office setups based on OSI layer, primary functions, and typical physical/logical ports.

1. Network Devices Comparison Table:

Network Device	Primary OSI Layer	Main Functions in Network Setup	Typical Physical Ports & Connectors	Typical Logical Ports / Protocols
Network Switch	Layer 2 (Data Link Layer)	Interconnects wired devices within the local LAN using	4 to 8 RJ45 Ethernet LAN Ports (10/100/	Layer 2 MAC Address Table lookup (No logical port

Network Device	Primary OSI Layer	Main Functions in Network Setup	Typical Physical Ports & Connectors	Typical Logical Ports / Protocols
		MAC addresses; forwards packets only to the specific destination port.	1000 Mbps).	management).
Router	Layer 3 <i>(Network Layer)</i>	Connects local LAN to the external Internet (WAN); performs IP packet	1 RJ45 WAN Port (Internet/ISP), 4 RJ45 LAN Ports , 1 Power	Port 80/443 (Web GUI Admin), Port 67/68 (DHCP Service),

Network Device	Primary OSI Layer	Main Functions in Network Setup	Typical Physical Ports & Connectors	Typical Logical Ports / Protocols
		routing, NAT translation, and DHCP IP assignment.	Jack, Reset Button.	Port 53 (DNS).
Hub	Layer 1 <i>(Physical Layer)</i>	Legacy multi-port repeater; broadcasts incoming data packets to ALL connected ports	4 to 8 RJ45 Ethernet Ports (Half-Duplex mode).	None (Operates purely at raw bit level with no frame filtering).

Network Device	Primary OSI Layer	Main Functions in Network Setup	Typical Physical Ports & Connectors	Typical Logical Ports / Protocols
		(causes high network collisions).		
Firewall	Layer 3 / 4 / 7 (Network to App)	Filters incoming and outgoing traffic based on security rules; blocks unauthorized access and	Dedicated WAN Inbound / LAN Outbound RJ45 Ports , USB Console Port.	Blocks/Allows TCP/UDP Ports (e.g., Port 80/443 HTTP/S, Port 22 SSH, Port 21 FTP).

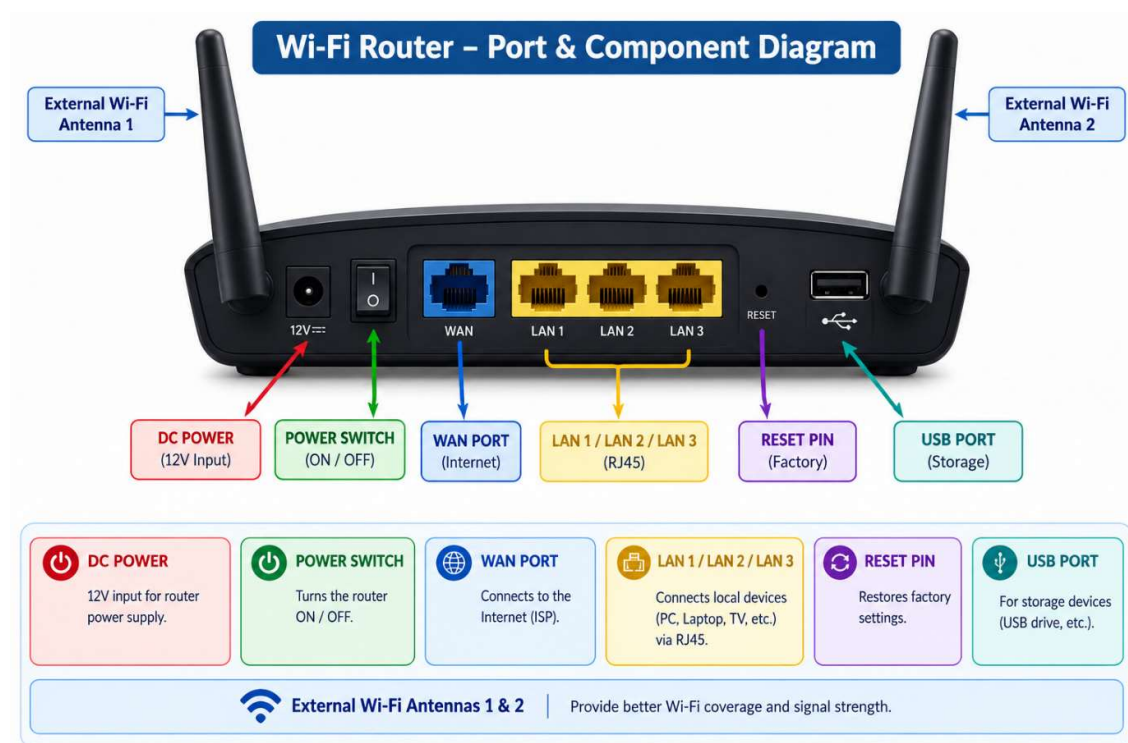
Network Device	Primary OSI Layer	Main Functions in Network Setup	Typical Physical Ports & Connectors	Typical Logical Ports / Protocols
		cyber threats.		
Access Point (AP)	Layer 1 / 2 <i>(Physical /Data Link)</i>	Converts wired Ethernet network signals into wireless 2.4 GHz/ 5 GHz Wi-Fi radio signals for mobile endpoints.	1 RJ45 Gigabit Ethernet Port (PoE/Uplink), External Wi-Fi Antennas, Power Jack.	Port 80/443 (AP Admin), Port 1812/1813 (RADIUS Authentication for WPA-Enterprise).

Task 2:

Network Devices Port Identification & Diagram Labeling

Objective: Illustrate physical port layouts and label critical physical interface ports (Ethernet LAN/WAN, Console, Power, Wireless Antennas, Reset) for 3 major network devices.

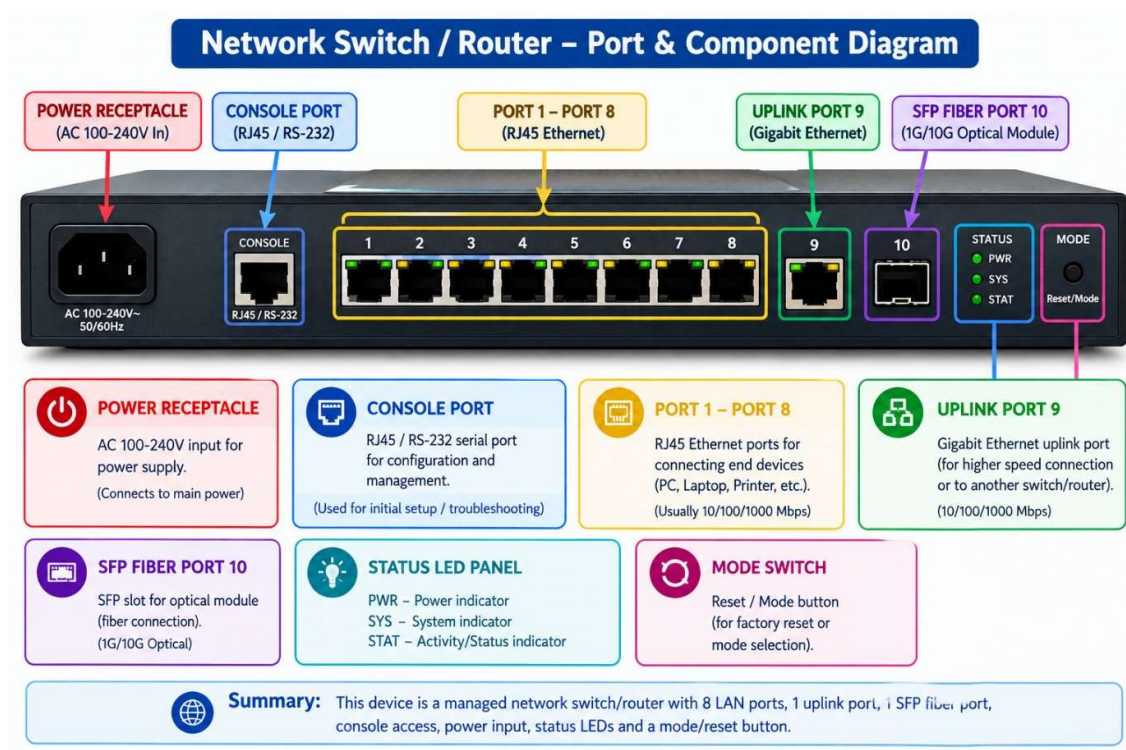
Device 1: Wireless Wi-Fi Router (Home / SOHO Router)



• Labeled Ports & Hardware Functions:

1. **Power Port (DC Jack):** Connects to the external power adapter (12V DC).
2. **WAN Port (Internet - Blue RJ45):** Connects to ISP Broadband Modem/Fiber terminal.
3. **LAN Ethernet Ports (Yellow RJ45):** 3 to 4 Fast/Gigabit Ethernet ports to connect wired PCs, Smart TVs, or Switches.
4. **Wireless Antennas:** Dual/Tri-band antennas transmitting 2.4 GHz and 5 GHz radio signals.

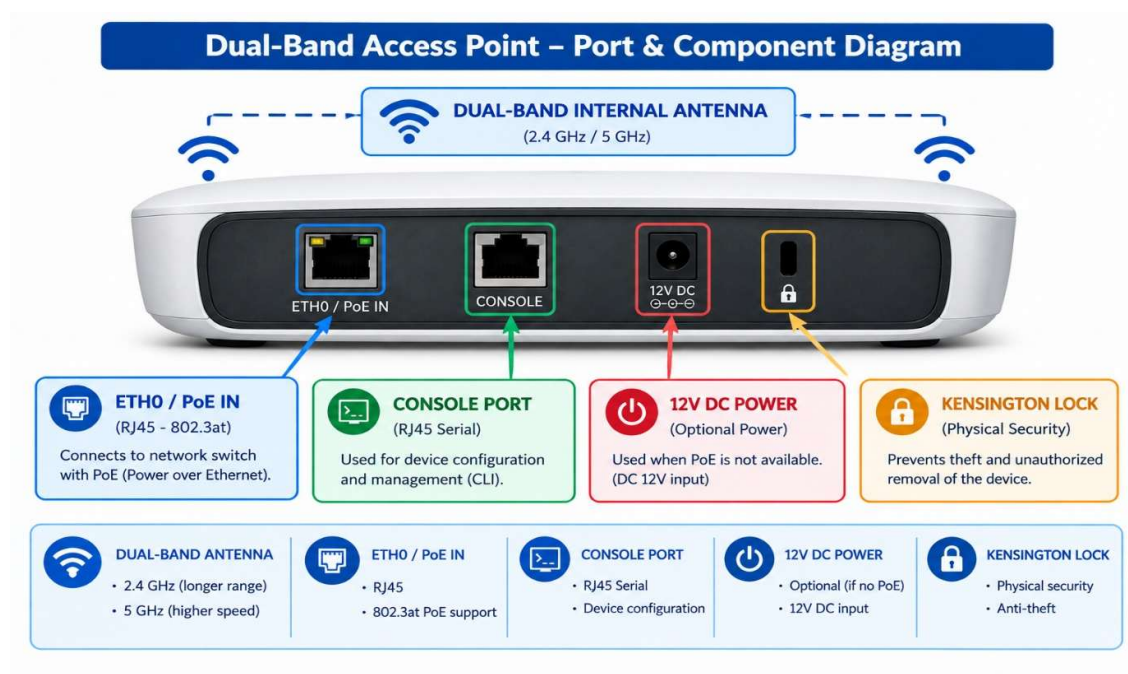
Device 2: Layer 2 Managed Network Switch



• Labeled Ports & Hardware Functions:

1. **Console Port (RJ45/DB9 Serial):** Used by network engineers to establish direct CLI session for configuration via serial terminal.
2. **Ethernet Access Ports (Ports 1–8):** High-speed RJ45 ports connecting end-user workstations, printers, and IP phones.
3. **SFP Fiber Uplink Port:** Accepts optical transceivers for long-distance high-speed fiber connectivity to core networks.
4. **AC Power Receptacle:** Internal power supply unit connector supporting standard 230V AC input.

Device 3: Enterprise Wireless Access Point (AP)



- **Labeled Ports & Hardware Functions:**

1. **Ethernet0 / PoE Input Port:** RJ45 Ethernet port receiving both network data and DC electricity via Power over Ethernet (PoE).
2. **Console Port:** RJ45 management port for localized command-line configuration.
3. **DC Power Jack:** Backup power input used when PoE power is unavailable.
4. **Internal Antenna Array:** Omnidirectional multi-element antenna array providing broad Wi-Fi coverage.

Task 3:

Real-World Application (Zomato) Network Device Interaction

Objective: Explain in 3-4 concise lines how specific network devices (Wireless Access Point, Router, and Firewall) handle network traffic when using a daily app like **Zomato**.

Application Role Description (Zomato App Example):

When I open **Zomato** to browse restaurants or place a food order, my smartphone first transmits radio signals to a **Wireless Access Point (AP)**, which converts the wireless Wi-Fi traffic into physical Ethernet frames.

Next, the local **Router** assigns a dynamic IP via DHCP and routes the API data packets across the internet to Zomato's cloud servers. Simultaneously, a **Firewall** monitors this traffic, inspecting incoming/outgoing data packets to allow secure SSL/TLS payment transactions while blocking potential unauthorized cyber threats.

Task 4:

College Wi-Fi Performance Troubleshooting Analysis

Objective: Diagnose the root cause of network slowness and random connection drops in a college campus Wi-Fi environment, and propose an hardware upgrade solution based on access point capacity, coverage, and traffic handling capabilities.

1. Identified Root Cause:

- **Primary Issue: Wireless Access Point (AP) Overload & Limited User Capacity**
 - **Detailed Reason:** College campus environments feature high client density (hundreds of simultaneous smartphone/laptop connections per hall). Legacy or consumer-grade Wireless Access Points can only handle a limited number of concurrent active connections (typically 30–50 devices). When student numbers exceed this threshold, the AP's processing queue becomes saturated, leading to co-channel interference, packet loss, bandwidth throttling (slowness), and forced client disconnections.
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2. Recommended Device Upgrade & Impact Solution:

- **Upgraded Hardware: Enterprise-Grade High-Density Wi-Fi 6 / 6E Access Points** (e.g., Cisco Catalyst / Aruba APs with MU-MIMO and OFDMA support).
- **How Upgrading the Access Point Resolves the Issue:**
 1. **Higher Client Capacity:** Enterprise Wi-Fi 6 APs can handle over 250 – 500 + concurrent connected client devices per unit without performance degradation.
 2. **Multi-User Traffic Handling (MU-MIMO & OFDMA):** Transmits data to multiple student devices simultaneously rather than sequentially, eliminating buffering and packet queuing delays.
 3. **Seamless Band Steering & Coverage:** Automatically steers devices from congested 2.4 GHz channels to faster 5 GHz/6 GHz bands, providing broader coverage and eliminating dead zones across campus.

